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International Mathematics Assessments for Schools

2023 / 2024 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2023 / 2024 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

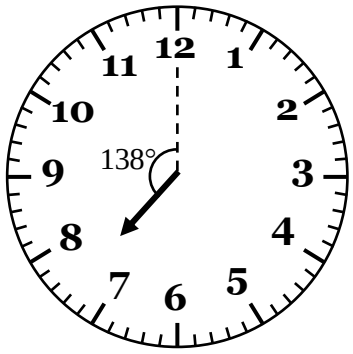
Questions 1-10, 3 marks each

1. How many integers are there between $\frac{210}{7}$ and $\frac{2}{3}\times 126$ inclusive?

（A）53　（B）54　（C）55　（D）56　（E）57

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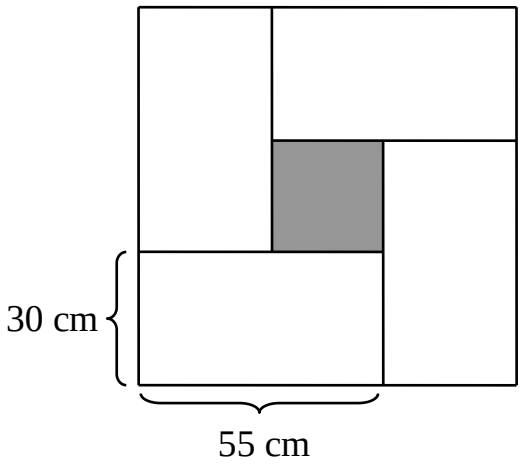
2. As shown in the diagram, the minute hand of a wall clock is missing. All the other accessories are normal. At a certain moment, the angle between the hour hand and 12 o'clock is measured to be 138°. What was the exact time of this moment?



（A）7:22　（B）7:24　（C）7:26　（D）7:28　（E）7:32

---

3. There are four identical rectangles arranged around a shaded square without overlapping, as shown in the diagram below. If the width of the rectangle is 30 cm and the length of the rectangle is 55 cm, what is the area, in cm^2, of the shaded square?



（A）625　（B）900　（C）1650　（D）2500　（E）3025

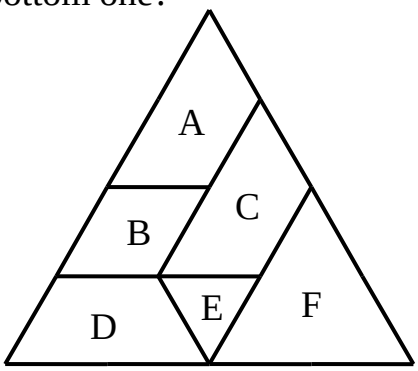
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4. If two identical machines can produce 180 items in one minute, how many items would five identical machines of the same type produce in 80 seconds?

（A）240　（B）360　（C）480　（D）600　（E）1200

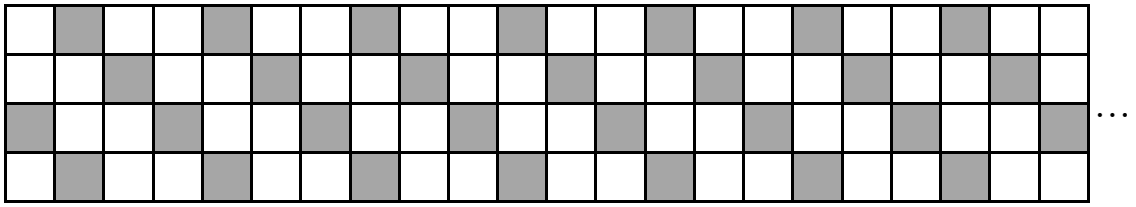
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5. Six identical equilateral triangle pieces, with the same direction, are piled on top of one another, resulting to the diagram shown below. Which is the fifth piece to be removed from the pile, if you remove the pieces in order, starting from the piece on the top to the bottom one?



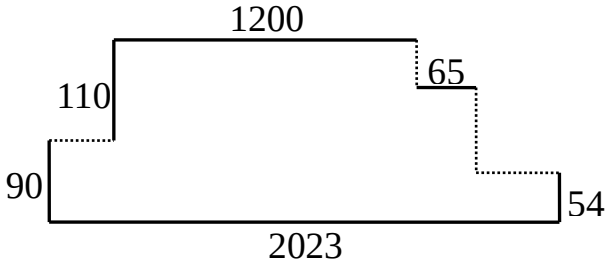
(A) A (B) B (C) C (D) D (E) E

6. Some unit squares of a 4×506 white grid are coloured in gray, resulting in some patterns, as shown in the diagram below. How many unit squares, out of the total of 2024 squares in this grid, are coloured in gray?



(A) 672 (B) 673 (C) 675 (D) 676 (E) 677

7. The diagram below shows a decagon where all its adjacent sides form a right angle. The lengths, in cm, of the solid sides are known and written on each side, while the lengths of the dotted sides are not known. What is the perimeter, in cm, of the decagon? Note: the diagram is NOT drawn to scale.

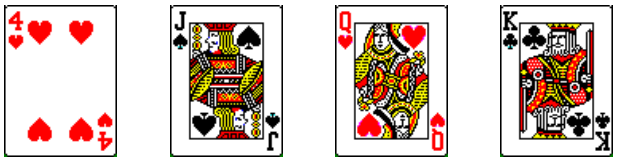


(A) 3542 (B) 3623 (C) 4046 (D) 4365 (E) 4446

8. Bob claims that he walks with a speed of 50 metres per minute. But in fact, he does not because he thinks that there are 60 centimetres in a metre and 100 seconds in a minute. What is his real walking speed?

(A) 15 metres per minute
(B) 18 metres per minute
(C) 25 metres per minute
(D) 27 metres per minute
(E) 50 metres per minute

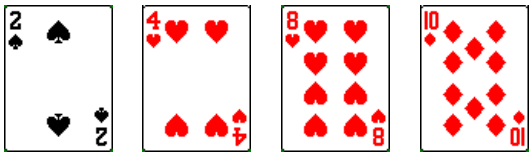
9. Four poker cards are randomly drawn from a deck. The goal is to combine the values of these cards by using the four arithmetic operators (+, −, × and ÷) to obtain 24. It is assumed that A, J, Q and K have the values of 1, 11, 12 and 13, respectively. For example, if the four cards are



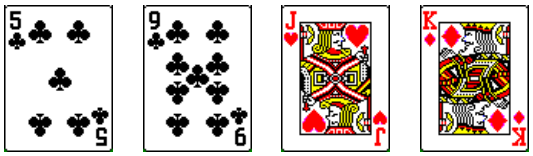
then one way to get 24 is $12 \div (13-11) \times 4 = 24$.

Now, there are five piles of cards, where four cards are in each pile:

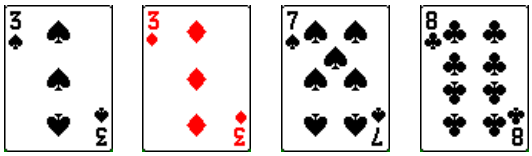
Pile (I)



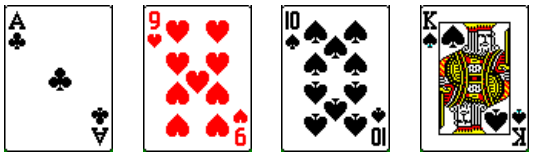
Pile (II)



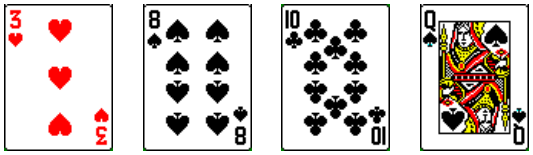
Pile (III)



Pile (IV)



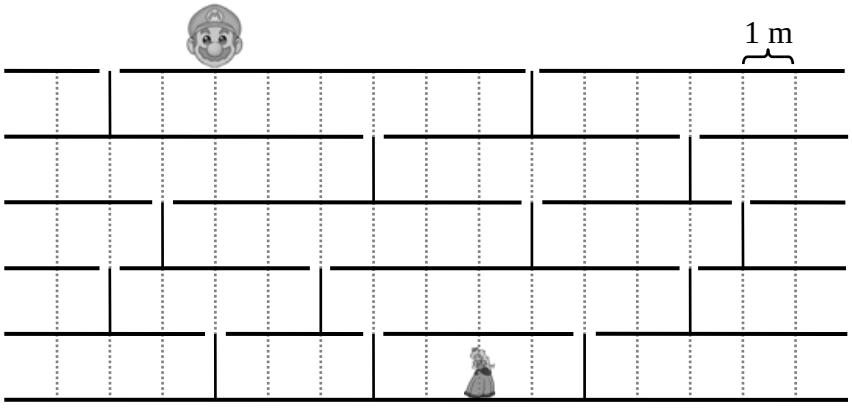
Pile (V)



For how many of these piles can get 24 by using the conditions stated above?

（A）5　　（B）4　　（C）3　　（D）2　　（E）1

10. Mario is on the top floor of a castle and wants to get to Princess Peach's location on the bottom floor. On each floor, there are holes with slider for him to slide down quickly without walking. But he needs to walk to the hole. What is the shortest distance, in m, that Mario has to walk? (The dashed lines are 1 m apart.)



（A）15　　（B）16　　（C）17　　（D）18　　（E）19

Questions 11-20, 4 marks each

11. Alice says that Brian is lying. Brian says that Chris is lying. Chris says that Alice and Brian are lying. Suppose a truth teller always tells the truth and a liar always tell lies. Who is telling the truth?

(A) Alice (B) Brian (C) Chris
(D) all three of them (E) none of them

---

12. The number of Brian's candies is 120% that of Alice's, the number of Brian's candies is 80% that of Colin's. Of what percent of the number of Colin's is that of Alice's?

(A) $33\frac{1}{3}\%$ (B) $66\frac{2}{3}\%$ (C) 80% (D) 100% (E) 150%

---

13. Which of the following numbers is a multiple of 7 and the sum of its digits is a prime number?

(A) 269 (B) 273 (C) 326 (D) 364 (E) 373

---

14. The Maths teacher chooses a non-negative integer and tells Alice this number, then tells Bob the sum of the number with 1, and tells Chris the sum of the number with 2. Which of the following numbers cannot be the product of these three numbers?

(A) 0 (B) 210 (C) 504 (D) 1716 (E) 2024

---

15. If 11 apples cost the same as 14 oranges, 22 strawberries cost the same as 21 oranges, 10 strawberries cost the same as 3 papayas and 5 peaches cost the same as 4 papayas, how many apples cost the same as 4 peaches?

(A) 8 (B) 12 (C) 16 (D) 20 (E) 24

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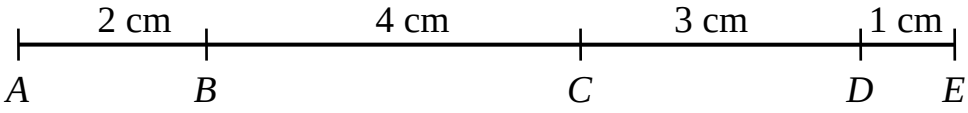
16. In the table below, every number starting from b equals 6 less than twice the sum of the two previous ones starting from the left. What is the value of $a+b+c+d$?

a	8	b	34	c	d

(A) 234 (B) 320 (C) 333 (D) 354 (E) 413

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17. There are marks on a wooden log that divide it into four sections, and their lengths are shown in the diagram below:

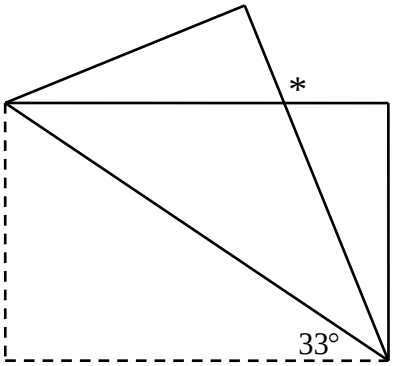


What is the total sum of the lengths, in cm, of all line segments with endpoints among the points A, B, C, D and E of this wooden log?

(A) 50 (B) 51 (C) 52 (D) 53 (E) 54

---

18. A rectangular piece of paper is folded along one of its diagonals, as shown in the diagram below. What is the measure, in degrees, of the angle marked with "*" ?



（A）110°　（B）113°　（C）114°　（D）119°　（E）120°

19. What is the largest positive integer less than 1000 which leaves a remainder of 5 when divided by 6, a remainder of 6 when divided by 7 and a remainder of 7 when divided by 8?

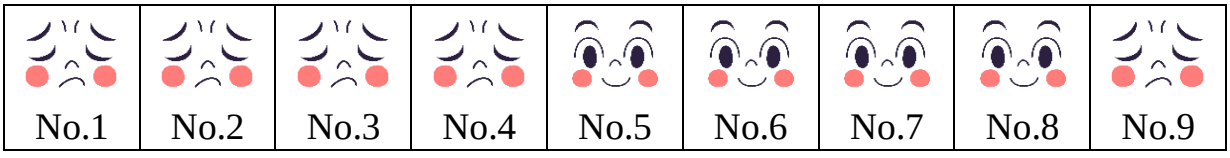
（A）167　（B）335　（C）671　（D）839　（E）907

20. Two friends, Andrew and Brad, travelled to an isolated village, when they got hungry. All they could buy from a bakery in that village was bread. One loaf of bread costs $1. Andrew bought 3 loaves of bread and Brad bought 2 loaves of bread. While they were about to eat, another person, Mr. X, came along asking for some food. The two friends decided to share all the bread with Mr. X, so that all three people would eat equally. When all the bread was eaten, Mr. X departed, leaving $5 to share between the two friends. At this time, the two friends got into an argument. Andrew argued that he must receive $3 for buying three loaves of bread, while Brad deserved only $2 for buying only two loaves of bread. So, in his view, the fair split of money would be $3 : $2. However, Brad claims that all three people shared the bread equally, so the fair split is $2.5 : $2.5.
What is the fair split ratio (Andrew : Brad) of the $5 between the two friends?

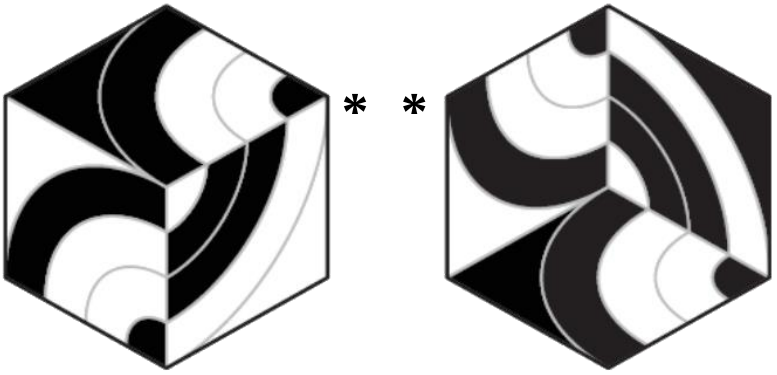
（A）$2 : $3　（B）$2.5 : $2.5　（C）$3 : $2
（D）$3.5 : $1.5　（E）$4 : $1

Questions 21-25, 6 marks each

21. Nine people, numbered 1 to 9 from left to right, stand in a row. Three of them always frown, three others always smile, while the remaining three mature people sometimes frown and sometimes smile. Each neighbour of a frowner is a mature person. At a certain moment, the frowns and smiles are shown in the diagram below. Use the numbers of three mature people from left to right to form a three-digit number. What is this three-digit number?

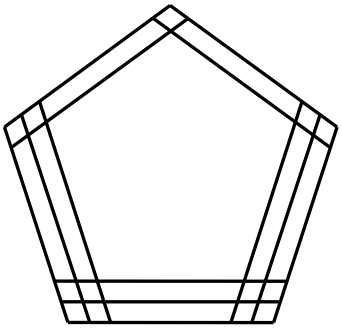


22. The surface of a cube is divided into black and white parts as shown in the diagram below. It is known that the side length of this cube is 4 cm, and on each face, the radius of the smallest quarter circle is 1 cm, the radius of second smallest quarter circle is 2 cm, the radius of second largest quarter circle is 3 cm and the radius of largest quarter circle is 4 cm. What is the sum of the areas, in cm^2, of all black regions? (The area occupied by the circular curve is negligible. Take $\pi = 3$)

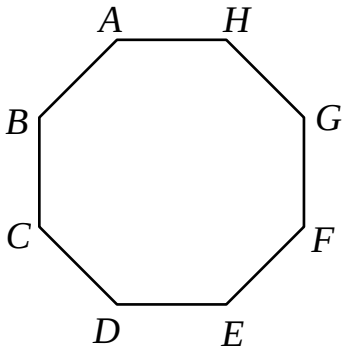


Top-front view Bottom-back view

23. How many pentagons can be observed in the diagram below?



24. In how many ways the given octagon *ABCDEFGH* can be divided into 3 quadrilaterals by drawing two nonintersecting diagonals?



25. The positive integers from 1 to 100 are written in a row. Each number is coloured with one colour, according to the following rule: if a number has a certain colour, then any of its proper divisors has the same colour. What is the maximum number of colours that can be used?

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ..., 98, 99, 100

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